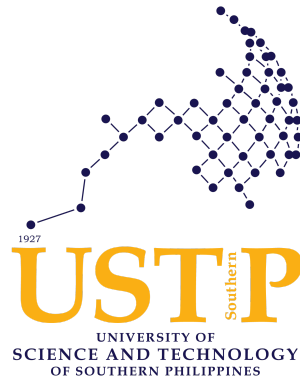


ChatDesk: An AI-Powered Mobile Inquiry System with Hybrid Chatbot and Ticket Escalation for Student Services at USTP



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TABLE OF CONTENTS

PAGE

ChatDesk: An AI-Powered Mobile Inquiry System with Hybrid Chatbot and Ticket Escalation for Student Services at USTP

CHAPTER I	1
1.1 Background of the Study	1
1.2. Statement of the Problem	5
1.2.1 General Problem	5
1.2.2 Specific Problems	5
1.3. Objectives of the Study	6
1.3.1 General Objective	6
1.3.2 Specific Objectives	6
1.4. Significance of the Study	7
1.5. Scope and Limitations	8
CHAPTER II	10
2.1. Preliminary Literature Review	10
2.2. Related Technologies	13
2.3 Software Tools	14
2.4 Published Research Articles, Journals, and Other Literature	15
2.5 Summary of Related Literature	16
CHAPTER III	17
3.1 Information Gathering	17
3.1.1 Interview with Key Stakeholders (Digital Transformation Office)	17

3.1.2 End-User Needs Analysis (Student Survey)	18
3.1.3 FAQ Knowledge Base Volume and Categorization	19
3.2 System Design	20
3.2.1 Development Model	20
3.2.2 System Architecture	21
3.2.3 Use Case Diagram	22
3.2.4 Context Diagram	24
3.2.5 Level 1 Data Flow Diagram	24
3.2.6 Level 2 Data Flow Diagram	26
3.3 Pilot Testing	38
3.4 System Performance Evaluation	41
3.4.1 System Usability Scale Testing	41
3.4.2 Evaluation Instrument	41
3.4.3 Manner of Conducting the Evaluation	43
3.4.4 Participants / Respondents of the Evaluation	44
3.4.5 Interpretation of Results Using the SUS Instrument	44
Reference	46

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The rapid advancement of digital technologies and artificial intelligence (AI) has significantly transformed the higher education sector worldwide. Educational institutions increasingly adopt digital platforms, automated systems, and AI-powered technologies to improve administrative efficiency, accessibility of services, and student engagement. According to Miao et al. (2021), digital innovation has the capability to “complement, enrich and transform education” by improving educational administration, governance, and communication systems within institutions. The authors further emphasized that digital technologies play an important role in strengthening accessibility and improving the quality of educational services globally.

In recent years, universities and colleges have also begun integrating AI-powered conversational systems such as chatbots and virtual assistants into their student support services. These systems are designed to automate repetitive inquiries, provide real-time responses, and reduce administrative workload. Studies have shown that AI-assisted systems improve communication efficiency and help institutions provide faster and more accessible support services to students. Furthermore, the growing adoption of AI technologies in higher education demonstrates the increasing need for institutions to modernize traditional inquiry and support mechanisms.

Several institutions and organizations worldwide have already implemented Artificial Intelligence (AI)-powered chatbot systems to improve communication efficiency and automate student support services. These conversational AI systems are designed to provide instant responses to frequently asked questions (FAQs), reduce administrative workload, and improve accessibility to institutional information. One example is Ivy.ai, an AI-powered chatbot platform widely used in higher education institutions. Ivy.ai provides automated assistance for admissions, enrollment, financial aid, scholarships, and campus-related concerns. According to Ivy.ai (2025), the platform enables universities to respond to student inquiries in real time while reducing repetitive workloads among staff members. The growing adoption of AI chatbots in universities

demonstrates the increasing demand for automated and accessible student support systems. Another example is Zendesk Answer Bot, an AI-driven automated response system that assists organizations in handling repetitive customer and student inquiries. The system uses machine learning and natural language processing (NLP) to identify user concerns and provide relevant responses automatically. Studies have shown that AI-assisted support systems improve communication efficiency and allow organizations to manage high inquiry volumes more effectively (Labadze et al., 2023). Building upon the proven effectiveness of these technologies, ChatDesk integrates advanced Large Language Models (LLMs), specifically utilizing the OpenAI API. Rather than relying on rigid keyword-matching scripts or building a custom NLP library from scratch, integrating OpenAI's natural language processing capabilities allows the proposed chatbot to dynamically understand student intent, process Taglish inquiries seamlessly, and map them to the university's approved knowledge base.

In the Philippine educational context, the University of the Philippines Open University developed the Ask Iska and IskOU chatbot systems to provide automated information support services for students and stakeholders. According to Serrano et al. (2021), the chatbot was implemented to improve accessibility to university information and reduce the burden on administrative personnel handling repetitive inquiries. The study found that chatbot systems can improve response efficiency and support continuous communication between universities and students. Research also supports the growing effectiveness of AI chatbot systems in higher education. According to Belda-Medina and Kokošková (2023), educational chatbots improve user interaction, accessibility, and satisfaction when designed with accurate and responsive communication mechanisms. Similarly, Assayed et al. (2024) emphasized that conversational AI systems significantly reduce administrative workload while improving student response time in academic advising and support services. Despite the effectiveness of these existing systems, several gaps remain. Many chatbot platforms are generalized enterprise solutions and are not specifically designed for localized university processes and procedures. Some systems also lack hybrid escalation mechanisms that allow unresolved concerns to be forwarded directly to human personnel. In addition, many universities in the Philippines still rely heavily on manual or ticket-based inquiry systems that do not provide real-time automated assistance.

In the Philippines, higher education institutions are continuously pursuing digital

transformation to improve the accessibility, efficiency, and quality of student services. The Commission on Higher Education (CHED) emphasized the importance of integrating digital technologies in higher education through its policies and strategic initiatives for flexible learning, digital governance, and institutional modernization. According to the Commission on Higher Education (2021), digital transformation enables universities to improve administrative operations, strengthen communication systems, and provide more accessible support services to students, particularly in increasingly technology-driven learning environments. The rapid adoption of digital platforms became more evident during and after the COVID-19 pandemic, when universities were compelled to shift many academic and administrative services online. The United Nations (2020) reported that educational institutions worldwide accelerated digital adoption to maintain continuity of services and communication during disruptions caused by the pandemic. In the Philippine context, this shift exposed the limitations of traditional manual processes and highlighted the need for more responsive and automated student support systems.

Despite these advancements, many Philippine universities still rely heavily on manual or semi-digital inquiry systems such as email-based support, social media announcements, online forms, and ticket-based helpdesks. While these methods provide some level of accessibility, they often depend on staff availability and manual responses, resulting in delays during peak periods such as enrollment, scholarship applications, and student organization activities. Students are also frequently required to visit offices physically to clarify procedures and requirements, contributing to congestion and operational inefficiencies within university support offices. Several studies have emphasized the growing need for AI-powered communication systems in Philippine higher education institutions. Serrano et al. (2021) explained that chatbot systems such as Ask Iska and IskOU were developed to improve institutional communication efficiency and provide automated information support to students. Their findings suggest that conversational AI systems can reduce repetitive inquiries and improve accessibility to university services.

Furthermore, the increasing use of mobile technologies among Filipino students creates an opportunity for institutions to implement mobile-first support platforms. Since students frequently rely on smartphones for communication and information access, integrating AI-powered inquiry systems into mobile applications can improve convenience, accessibility,

and responsiveness in delivering student services. These conditions demonstrate that while Philippine higher education institutions are gradually adopting digital systems, gaps in real-time automated student support still exist. This highlights the need for localized AI-powered solutions that can improve communication efficiency, reduce administrative workload, and enhance the overall student experience within universities.

The University of Science and Technology of Southern Philippines (USTP) currently provides an online Helpdesk system through its official website to assist students with inquiries and concerns. The system mainly functions as a ticket-based platform where students submit concerns that are later addressed by assigned personnel. While this reduces the need for physical visits, the process still relies heavily on manual responses and staff availability. As a result, students may experience delays in receiving information, especially during peak periods such as enrollment, scholarship processing, and campus activities. The current system also lacks real-time automated responses for frequently asked questions, causing repetitive inquiries to continuously interrupt staff workflow. In addition, students often rely on fragmented sources of information such as social media pages, emails, or physical office visits to clarify procedures and requirements.

Although USTP continues to pursue institutional modernization through its Vision 2030 initiatives, the university still lacks a dedicated AI-powered inquiry system capable of providing instant responses, automated guidance, and intelligent inquiry management for student services. These limitations highlight the need for a localized hybrid chatbot system such as ChatDesk to improve communication efficiency, reduce operational congestion, and enhance the overall student support experience within the university.

1.2 Statement of the Problem

1.2.1 General Problem

The general problem of this study is the limitation of the existing online Helpdesk system of the University of Science and Technology of Southern Philippines (USTP) in efficiently handling student inquiries across the university. The current system mainly operates as a ticket-based platform, which relies on manual responses from personnel and lacks real-time interaction. As a result, students still experience delays in receiving information and continue to rely on physical offices or fragmented online sources. Additionally, the system does not effectively handle repetitive inquiries automatically, leading to frequent interruptions in staff workflow. These limitations result in inefficiencies, slower service delivery, and difficulty in managing high volumes of student concerns, especially during peak periods.

1.2.2 Specific Problems

A.) Prolonged Student Waiting Times for Routine Information:

Students experience excessive waiting times at Student Affairs service counters when seeking guidance on procedures such as scholarship applications, counseling appointments, student organization registration, ID services, campus event participation, or academic FAQs. This contributes to campus congestion and student frustration.

B.) Decreased Staff Productivity Due to Repetitive Inquiries :

Personnel in the Student Affairs Office and related support services are disrupted by continuous repetitive inquiries. These interruptions delay their core responsibilities, such as coordinating student welfare programs, managing events, maintaining student records, and providing quality guidance.

C.) Absence of a Centralized Interactive AI Assistant:

Despite the availability of online portals and informational resources, there is no dedicated,

interactive chatbot application that can filter and respond to dynamic FAQs for Student Affairs and related academic guidance. Students are forced to rely on physical inquiries, which is inefficient and time-consuming.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to develop and implement **ChatDesk: An AI-Powered Mobile Inquiry System with Hybrid Chatbot and Ticket Escalation for Student Services at USTP**. This system aims to automate the dissemination of frequently asked questions and routine procedural information, thereby reducing operational bottlenecks, minimizing physical queuing times for students, and improving overall staff productivity by allowing personnel to focus on more complex tasks related to student support and welfare.

1.3.2 Specific Objectives

1. To design and develop an interactive hybrid chatbot application utilizing React Native and the OpenAI API to provide instant, natural language responses and procedural guidance for routine student inquiries related to the Office of Student Affairs and Services (OSA).
2. To implement a centralized, web-based administrative dashboard and a Django-powered ticket escalation backend that enables OSA personnel to efficiently manage the FAQ knowledge base and manually resolve complex student concerns.
3. To test and evaluate the functionality, usability, and acceptability of the system using the System Usability Scale (SUS) instrument.

1.4 Significance of the Study

The USTP Office of Student Affairs and Services: The implementation of **ChatDesk** will significantly enhance the operational efficiency of the Offices of Student Affairs by automating the handling of routine and repetitive student inquiries. By filtering common questions through an AI-driven system, staff members are shielded from constant interruptions, allowing them to focus on more complex tasks related to student support, welfare, and campus activities. Furthermore, the system provides valuable analytics on the most frequent student concerns such as scholarship processes, counseling requests, ID services, and student organization inquiries enabling the office to make data-driven decisions to improve service delivery. This transition to a digital information hub modernizes the office's operations and aligns with the university's digital transformation goals.

The Students: Students will benefit from a more streamlined and accessible way to obtain vital university information without the need to travel to campus or wait in long physical queues. The ChatDesk application provides instant responses to procedural questions, ensuring that students can plan their academic requirements at their own convenience. This immediate access to accurate information reduces the frustration associated with bureaucratic delays and ensures that all students receive consistent and standardized guidelines. Ultimately, the system improves the overall student experience by making administrative interactions faster, clearer, and more efficient.

Future Researchers (Others) This study will serve as a valuable reference and foundation for future IT researchers interested in conversational AI and university administrative automation. The methodologies and system architecture documented in this research can be used as a blueprint for developing similar chatbot systems in other departments within the university or in other educational institutions. Additionally, the findings of this study provide insights into user acceptability and the effectiveness of AI in a local academic context. It opens up opportunities for further exploration into more advanced Natural Language Processing (NLP) applications for campus-wide services.

1.5 Scope and Limitations

Scope of the Study:

The primary focus of this study is the development of **ChatDesk**, An AI-Powered Mobile Inquiry System with Hybrid Chatbot and Ticket Escalation for Student Services at USTP CDO campus. The system will handle frequently asked questions related to student services such as scholarships, counseling, student organizations, and campus activities, as well as college-related concerns including enrollment procedures, subject advising, and shifting processes.

The data used in the system will be gathered from official procedures, frequently asked questions (FAQs), and announcements provided by the respective offices within USTP CDO. The primary users of the system include students and authorized personnel who will manage and update the information. To access the mobile application, students are required to log in using an email address (preferably their official university email). This Single Sign-On (SSO) approach allows the system to link unhandled or complex inquiries to the specific student, enabling the chatbot to escalate tickets directly to the appropriate personnel for manual review. For implementation purposes, the system will be piloted and evaluated within selected offices under the Office of Student Affairs and Services

Limitations of the Study:

- **No Integration with Student Academic Records:** The system will not connect to the PRISMS database. It cannot display individual student grades, remaining balances, or personal subject schedules.
- **No Transaction Processing:** The system is strictly for inquiries. It cannot process actual document payments, submit shifting forms, or register students into subjects.
- **Limited Data Handling for Privacy Compliance:** While the system utilizes an email-based login to facilitate the ticketing and escalation process, it restricts data

collection to basic profile information (email and name). This ensures compliance with the Data Privacy Act of 2012 by collecting only the necessary data for administrative follow-ups without exposing highly sensitive academic records.

- **Dependency on Internet Connectivity:** As a mobile application, the system requires an active internet connection to function and cannot be accessed offline.
- **Limited Language Support:** The chatbot will be designed to process and respond primarily in English and Tagalog (or "Taglish"), which are the common mediums of communication on campus.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of relevant literature, existing systems, and underlying technologies that provide the foundational framework for this study. The reviewed materials focus on the integration of artificial intelligence in educational frontline services, the effectiveness of mobile-first communication tools, and the mechanics of hybrid ticketing systems. These concepts validate the development of **ChatDesk**, an AI-Powered Mobile Inquiry System with Hybrid Chatbot and Ticket Escalation designed to streamline student inquiries at the University of Science and Technology of Southern Philippines (USTP), supported by a web-based administrative dashboard for frontline staff.

2.1 Existing Systems and Applications

The development of automated communication tools has seen widespread adoption across various industries, including educational institutions. Several existing platforms and frameworks currently dominate the conversational AI landscape:

OpenAI Chatbot Platform: Provides conversational AI assistants capable of natural language understanding and context-aware responses, improving automated communication and scalable student support (Chukwuere, 2024).

Google Dialogflow: A chatbot development framework that uses Natural Language Processing (NLP) for intent detection and multilingual interaction, commonly used in university inquiry systems (Belda-Medina & Kokošková, 2023).

IBM Watson Assistant: An enterprise AI chatbot platform used to automate administrative tasks such as answering frequently asked academic questions (Labadze et al., 2023).

Microsoft Azure Bot Service: A cloud-based chatbot deployment platform that enables scalable and secure student support systems accessible through web and mobile applications (Hsain & El Housni, 2024).

Amazon Alexa: A voice-enabled assistant integrated into educational environments to improve accessibility and student engagement through hands-free interaction (Iftikhar & Shah, 2025).

University Academic Advising Chatbot Systems: Institutional chatbot implementations used for automating enrollment, registrar inquiries, course requirements, and document procedures (Assayed et al., 2024).

Moodle Chatbot Plugin Model: A learning management system integration that allows automated responses within course portals to support student inquiries (Okonkwo & Ade-Ibijola, 2021).

Zendesk Answer Bot: A prominent enterprise customer service AI that utilizes a hybrid architecture. It attempts to resolve user inquiries using a pre-established knowledge base and automatically escalates unresolved interactions into formal support tickets for human agents. This directly mirrors the hybrid escalation bridge designed for the ChatDesk system.

Ivy AI: A conversational AI chatbot platform designed specifically for higher education institutions. It automates student support across various university departments, such as admissions and registrar services, demonstrating the effectiveness of institutional knowledge bases in reducing frontline administrative workload.

Iska & Isko (University of the Philippines): A localized, Philippine-based university chatbot designed to assist UP students with campus-specific inquiries, enrollment FAQs, and university guidelines. The success of this system highlights the growing demand and acceptability of localized, institution-specific conversational agents within the Philippine educational landscape.

Erica (Bank of America): An advanced virtual assistant that demonstrates high-accuracy intent classification and the secure handling of user-specific data. It serves as an industry benchmark for how conversational AI can reliably process complex, real-world queries in a highly regulated institutional environment.

USTP Helpdesk : The current student inquiry and support workflows at the University of Science and Technology of Southern Philippines (USTP) rely predominantly on a manual, online ticketing and digital messaging system. While this online setup provides direct staff assistance, it lacks an automated, initial AI filtering mechanism. This limitation requires frontline personnel to manually process every incoming query, which often leads to administrative fatigue, delayed response times, and overwhelming digital ticket backlogs during peak academic

periods—operational challenges that the intelligent, hybrid architecture of ChatDesk directly aims to resolve.

Comparison of Existing Systems:

Features	Proposed System	Ivy AI	Zendesk Answer Bot	Erica	Iska & Iskou	USTP Helpdesk
Basic Analytic Dashboard	✓	✓	✓	✓	X	✓
Announcement Broadcast Module	✓	✓	X	✓	X	X
AI-Powered Student Inquiry Handling	✓	✓	✓	X	✓	X
Intent Classification with basic NLP	✓	✓	✓	✓	✓	X
Quick Action Buttons	✓	✓	✓	✓	✓	✓
Automated Ticketing & Escalation	✓	✓	✓	✓	✓	✓
Admin Knowledge Base & Content Management Panel	✓	✓	✓	✓	X	✓
Phase Aware Student Guidance	✓	X	X	X	X	X

Synthesis and Justification of the Proposed System

While the aforementioned systems and frameworks provide robust natural language processing capabilities, they are largely generalized enterprise solutions or broad educational plugins. Platforms like Google Dialogflow and IBM Watson provide the infrastructure for AI, and traditional platforms like Zendesk offer ticketing, but they do not solve the localized operational bottlenecks at the University of Science and Technology of Southern Philippines (USTP). Furthermore, while localized bots like UP's Iska & Isko exist, they are proprietary to their respective institutions. ChatDesk distinguishes itself by utilizing robust NLP capabilities and embedding them into a newly localized, hybrid ecosystem. Unlike generalized chatbots, ChatDesk provides a dedicated mobile-first frontend specifically tailored for USTP students, combined with a seamless web-based ticketing system that escalates unresolved, complex inquiries directly to human OSA administrators, ensuring no student concern is left unanswered.

2.2 Related Technologies

The Shift to Mobile-First Campus Communication

The reliance on physical frontline offices for routine university procedures often leads to operational bottlenecks and prolonged student queuing. To address this, modern institutional frameworks are heavily migrating toward mobile-based solutions. Mobile applications provide students with instantaneous, on-the-go access to information, eliminating the constraints of traditional office hours. By deploying ChatDesk as a native mobile application, the system directly aligns with the digital habits of the modern student body, ensuring that procedural guidelines, enrollment steps, and general inquiries are accessible 24/7 directly from their smartphones.

Artificial Intelligence and Hybrid Chatbot Systems

While traditional helpdesks rely entirely on human intervention, modern AI-powered chatbots utilize Natural Language Processing (NLP) and generative Large Language Models (LLMs) to autonomously classify user intent and provide immediate, pre-approved responses to frequently asked questions (FAQs). In the implementation of ChatDesk, this NLP capability is driven by the integration of the OpenAI API. Utilizing this existing LLM infrastructure allows

the system to achieve advanced semantic understanding of dynamic student queries, including Taglish, without the need for manually constructing custom keyword dictionaries or training models from scratch. However, fully automated systems often struggle with highly complex or unique user concerns. To mitigate this, ChatDesk employs a "hybrid" architecture. As supported by recent studies on generative AI chatbots in higher education (Chukwuere, 2024), hybrid systems maximize efficiency by allowing the AI to filter and resolve routine questions, while seamlessly escalating unresolved issues as support tickets to a human administrator. This ensures that the system acts as a protective filter for the Office of Student Affairs and Services (OSA), reducing administrative fatigue while maintaining high-quality student support.

2.3 Software Tools

The successful deployment of a hybrid chatbot system requires a robust, dual-platform architecture that caters to both the end-user (students) and the system administrators (OSA staff). The development of ChatDesk integrates the following technological stack:

Mobile Frontend Development (React Native): To ensure maximum accessibility for the entire student body, the ChatDesk mobile application is developed using React Native. This cross-platform framework allows the chatbot interface to be deployed simultaneously as a native application on both Android and iOS (iPhone) devices from a single JavaScript codebase. By utilizing React Native, the system ensures a seamless, native-like user experience while significantly reducing development time and maintaining consistency with the ReactJS-based administrative dashboard.

Web-Based Administrative Dashboard (ReactJS): While students require mobile accessibility, frontline staff require an expansive interface for heavy data entry and analytics monitoring. Therefore, the Admin Panel is developed as a web application using ReactJS. This allows OSA personnel to efficiently manage the knowledge base, monitor chat metrics, and resolve escalated tickets using desktop computers within their office environment.

Natural Language Processing (OpenAI API): The core intelligence of the chatbot relies on a cloud-based Natural Language Processing (NLP) engine to parse Taglish

(Tagalog-English) inquiries. For this, ChatDesk utilizes the OpenAI Application Programming Interface (API) to access advanced Large Language Models (LLMs). The API analyzes and classifies the intent behind student messages, extracting context and keywords. By accurately mapping the student's intent, the system can fetch the correct pre-approved responses from the knowledge base or automatically trigger a ticket escalation if human intervention is required.

Backend and Database Management (Django REST Framework, DigitalOcean, & PostgreSQL): The backend of the ChatDesk system is developed using the Django REST Framework, which provides secure and scalable API services for handling user authentication, ticket routing, and cross-platform communication. To guarantee high availability, this backend is deployed on DigitalOcean, ensuring the chatbot and ticketing system remain online 24/7 independently of local university servers. Furthermore, PostgreSQL serves as the central cloud database for securely storing student accounts, FAQ entries, chat logs, and ticket records. It was specifically selected for its advanced ability to efficiently process the complex JSON data generated by AI interactions and its superior capacity to handle simultaneous database requests during peak enrollment periods.

2.4 Published Research Articles, Journals, and Other Literature

Several scholarly works support the implementation of chatbot systems in educational environments.

Assayed et al. (2024) conducted a systematic review of conversational AI chatbots used in academic advising and found that automated assistants significantly reduce administrative workload while improving student response time. Belda-Medina and Kokošková (2023) introduced the Chatbot-Human Interaction Satisfaction Model (CHISM), which evaluates user satisfaction in educational chatbot systems. Their findings emphasize usability and response accuracy as key success factors.

Labadze et al. (2023) analyzed the role of AI chatbots in education and concluded that chatbot integration enhances student engagement and accessibility to institutional services. Chukwuere (2024) proposed a conceptual framework for generative AI chatbots in higher education, highlighting their potential for automating student support services. Hsain and El

Housni (2024) discussed the use of large language model chatbots in international student support systems and found that conversational automation improves institutional communication efficiency. Iftikhar and Shah (2025) examined the relationship between AI chatbots and student engagement, concluding that conversational systems encourage active digital interaction.

These studies collectively support the relevance of developing a mobile-first, hybrid chatbot system for student inquiry automation.

2.5 Summary of Related Literature

The reviewed literature and systems demonstrate that AI-powered chatbot platforms are widely used in educational institutions to automate routine inquiries and improve service efficiency. Existing applications highlight features such as automated FAQs, natural language interaction, cloud deployment, and interactive user interfaces. Supporting technologies such as Artificial Intelligence and Natural Language Processing enable intelligent communication, while development tools such as React Native, ReactJS, Node.js, and MySQL provide the technical framework for a scalable, cross-platform implementation. These findings strongly support the development of the **ChatDesk** system as a modern, hybrid solution for automating student inquiry services at the university level.

CHAPTER 3 METHODOLOGY

3.1 Information Gathering

3.1.1 Interview with Key Stakeholders (Digital Transformation Office)

Background and Current Operations

To establish a baseline for the ticketing system's architecture, an interview was conducted with personnel from the Digital Transformation Office (DTO), who manage the university's current technical help desk. During regular, non-peak periods, the DTO help desk receives an average of 20 online tickets per day, as most students opt to visit the office in person for their individual concerns. However, ticket volume increases significantly during peak academic seasons, specifically during enrollment, enrollment adjustments, and LCRP examinations. Currently, the DTO utilizes a Facebook Page and Messenger to handle basic, automated queries, such as providing default system usernames and passwords. For issues requiring further action, such as account resets due to forgotten passwords, students are advised to submit a formal support ticket for manual assistance.

Problem Validation and Challenges

The interview identified specific operational bottlenecks in the current ticket resolution process. The most common and repetitive inquiries submitted involve the activation and deactivation of university academic system accounts, including PRISMS, USTeP, and eClearance. While personnel can typically resolve these tickets within the same day, assuming no backend system errors occur, they face significant challenges regarding how students submit their issues. The primary causes of delayed resolutions are the submission of duplicate tickets by the same student and the failure to provide sufficient context or information regarding their concerns. This lack of initial information necessitates prolonged back-and-forth conversations, preventing the staff from resolving the issue efficiently within a single ticket interaction.

Capability to Adapt the System and Feature Requirements

The insights gathered from the DTO validate the integration of ChatDesk's proposed features. The personnel confirmed that implementing an AI-powered chatbot would highly benefit support workflows by instantly answering common concerns and filtering out basic, repetitive issues before they require human escalation. Furthermore, the interview revealed a lack of comprehensive data analytics in the current setup; the office does not generate detailed ticket statistics or document ticket histories, relying solely on manually tracking the daily count of resolved tickets. Crucially, to improve the current help desk process, the DTO personnel explicitly requested a duplicate ticket detection or reminder system. Implementing an auto-tagging feature or an automated prompt reminding students to check their existing tickets would significantly reduce the burden of repeated submissions and streamline administrative support operations.

3.1.2 End-User Needs Analysis (Student Survey)

Problem Validation (Current System Inefficiencies)

To validate the current challenges faced by the student body, an online survey was conducted with 203 respondents from USTP. The data strongly highlights the inefficiencies of the current inquiry process:

- **Over-reliance on Walk-ins:** A significant majority of respondents (180 out of 203, or 88.7%) indicated that "Going physically to the office (Walk-in)" is their primary method of resolving university-related questions.
- **Delayed and Inconsistent Online Replies:** When messaging offices online, 39 students reported waiting 1 to 2 days for a clear answer, and an additional 11 students stated they "rarely/never get a reply"
- **Core Frustrations:** The most dominant frustrations cited by the student body revolve around "Long physical lines at the office," often compounded by "Slow reply times online" and "Offices being closed during free time".

Needs Analysis (Validating ChatDesk's Features)

The responses provide concrete, quantitative evidence that the specific features developed for ChatDesk directly address the student body's needs:

- **High Demand for AI Availability:** Students are highly receptive to an automated solution. When asked how likely they are to use a 24/7 AI chatbot instead of waiting for human staff, the average response was an overwhelming 4.77 out of 5, with 162 students giving it a perfect score.
- **Validation for the Taglish NLP Engine:** The decision to implement an NLP engine capable of parsing Taglish is strongly supported. 152 out of 203 students (74.9%) stated that a system allowing them to type in "Taglish" would "definitely" make them more comfortable using it.
- **Validation for Hybrid Ticket Escalation:** The hybrid escalation model is critical to the end-users. When asked how important it is for the system to automatically forward complex questions to real human staff as a "Support Ticket," the average score was 4.85 out of 5, with 180 students rating it as highly important.
- **Platform Accessibility (Mobile App Preference):** The dual-platform architecture (React Native mobile app for students) is strongly validated, as 139 respondents (68.5%) strictly prefer accessing university services via a dedicated Mobile App over a web browser.
- **Targeted Knowledge Base (FAQs):** Manual text inputs revealed that students consistently struggle to find answers regarding Clearance processes, Enrollment, and Scholarships, indicating the precise topics that must be prioritized in the chatbot's initial knowledge base training.

3.1.3 FAQ Knowledge Base Volume and Categorization

To fully justify the integration of a Large Language Model (LLM) and ensure the system's practical utility, the study will gather and integrate the entirety of the official frequently asked questions (FAQs) provided by the Office of Student Affairs and Services (OSA). Capturing the complete operational knowledge base of the OSA inherently guarantees a

substantial and highly complex volume of data. This extensive, department-wide dataset encompasses all core categories, including enrollment procedures, scholarship and financial aid guidelines, student clearance requirements, recognized student organization (RSO) protocols, and campus disciplinary violations. Managing the entirety of this highly specific and frequently overlapping institutional data makes traditional, simple database queries highly inefficient and prone to error. Deploying the OpenAI API is therefore a structural necessity; the LLM is required to accurately perform semantic intent classification across this massive array of dynamic data points. This ensures the system can precisely differentiate between closely related inquiries—such as an academic violation versus a clearance hold—guaranteeing that students receive the exact policy information they need without generating false positives.

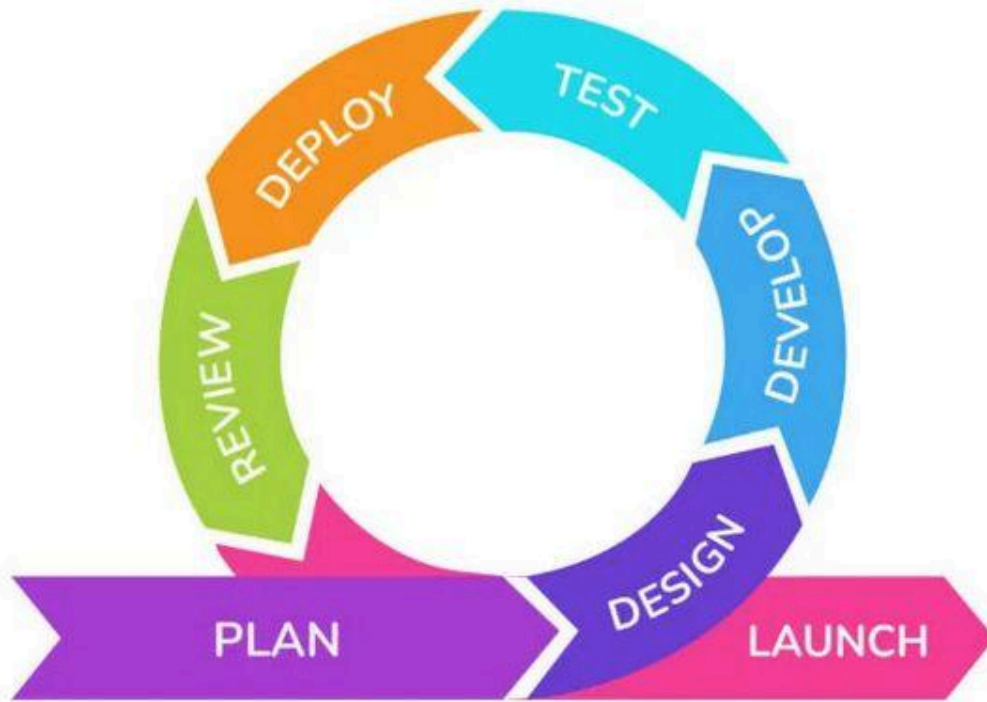
3.2 System Designs

3.2.1 Development Model

Development Model Used: The development of the ChatDesk system utilized the Agile Development Model, which prioritizes adaptive planning and continuous improvement (Beck et al., 2001). This model was selected due to its flexibility and iterative approach, which is highly effective for cross-platform and mobile application development where system requirements frequently evolve (Alsaqqa et al., 2020), particularly in refining the chatbot's Natural Language Processing (NLP) accuracy for Taglish queries and optimizing the automated ticket escalation workflow for OSA administrators.

3.2.2

AGILE



System Architecture

The **ChatDesk** system employs a dynamic **3-Tier Architecture** designed to ensure secure and scalable operations while clearly isolating the artificial intelligence processing. The Presentation Layer serves as the dual user interface, utilizing React Native for the student mobile application and ReactJS for the staff's web-based admin panel. All core business logic is processed in the Application Layer, where a Django REST Framework backend hosted on DigitalOcean acts as the central API gateway. To handle natural language processing efficiently, the system's architecture deliberately separates the AI component from the core backend. When a student submits a query, Django securely interacts with the external OpenAI API, which is solely responsible for classifying student intent before handing the data back to the core engine. Finally, the Data Layer acts as the conceptual storage hub, utilizing a PostgreSQL relational database for

the secure, persistent storage of user accounts, the chatbot's knowledge base, automated ticket records, and system analytics.

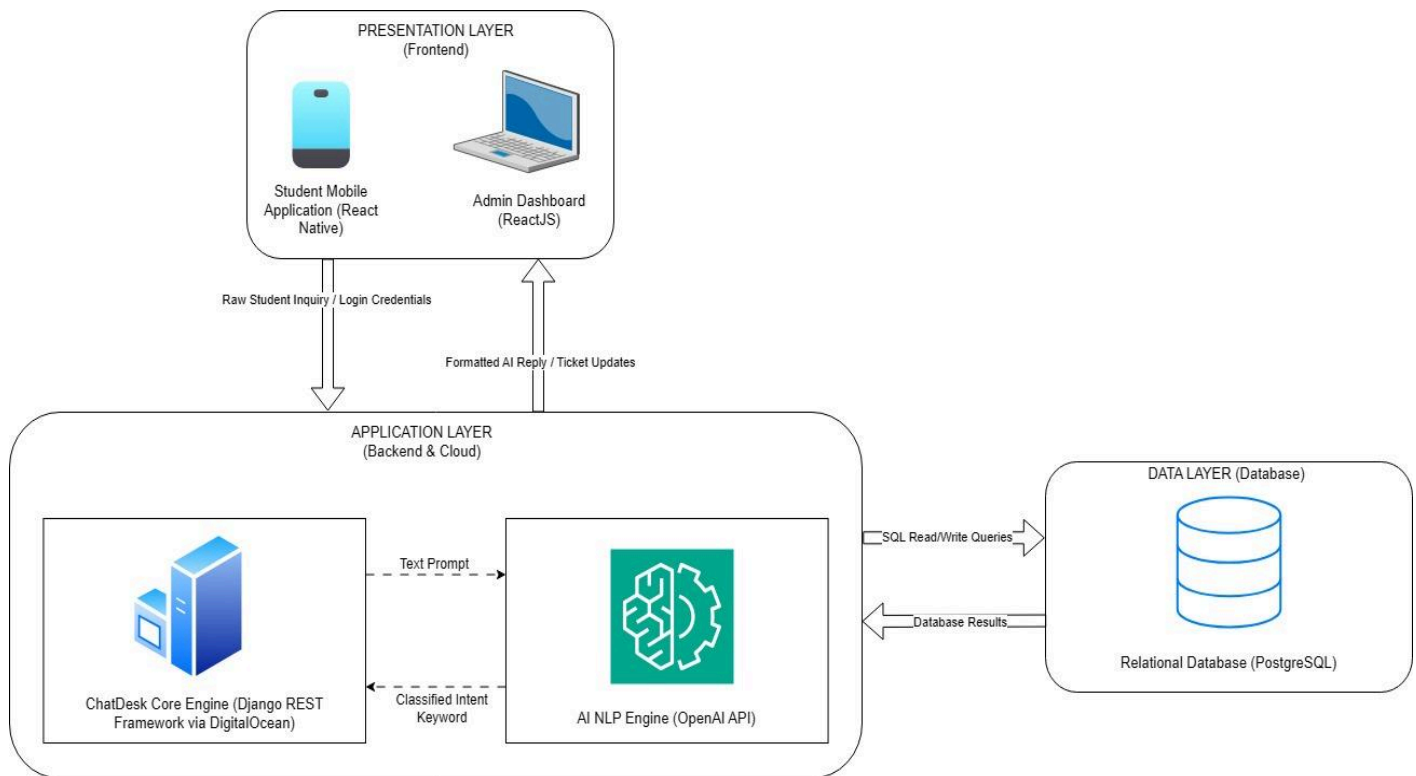


Figure 1. System Architecture of ChatDesk

3.2.3 Use Case Diagram

The Use Case Diagram illustrates the interactions between the ChatDesk system and its two primary actors: the **Student** and the **Admin/Staff**.

The **Student** acts as the end-user who can securely log in, send inquiries to the AI chatbot, and track ticket statuses. If the chatbot cannot answer a complex question, the system uses an <<extend>> relationship where it automatically generates a support ticket on behalf of the student, immediately forwarding the concern to the OSA staff for human intervention.

The **Admin/Staff** manages the backend helpdesk operations. They can monitor chat analytics, update the AI's FAQ knowledge base, and resolve escalated support

tickets. When an admin resolves a ticket, the system strictly requires (<<include>>) to update the ticket status and automatically notify the student of the resolution. Additionally, the admin has the optional (<<extend>>) ability to instantly save the solution as a new FAQ, allowing the chatbot to continuously learn from manual interactions.



3.2.4 Context Diagram

The Context Diagram illustrates the overarching data flow between the central ChatDesk system and its two primary external actors: the Student and the Admin/Staff. Students interact with the platform by providing their login credentials for authentication and submitting student inquiries directly to the AI interface. In return, they receive immediate response and guidance from the automated chatbot, alongside real-time ticket status updates for any complex issues routed to human support.

On the administrative side, the Admin/Staff interact with the system by supplying continuous FAQ updates to maintain the AI's knowledge base and providing direct resolutions to automatically escalate support tickets. In return, the system provides administrators with these auto-generated tickets that require human intervention, as well as comprehensive system analytics to effectively monitor overall ChatDesk performance and inquiry trends.

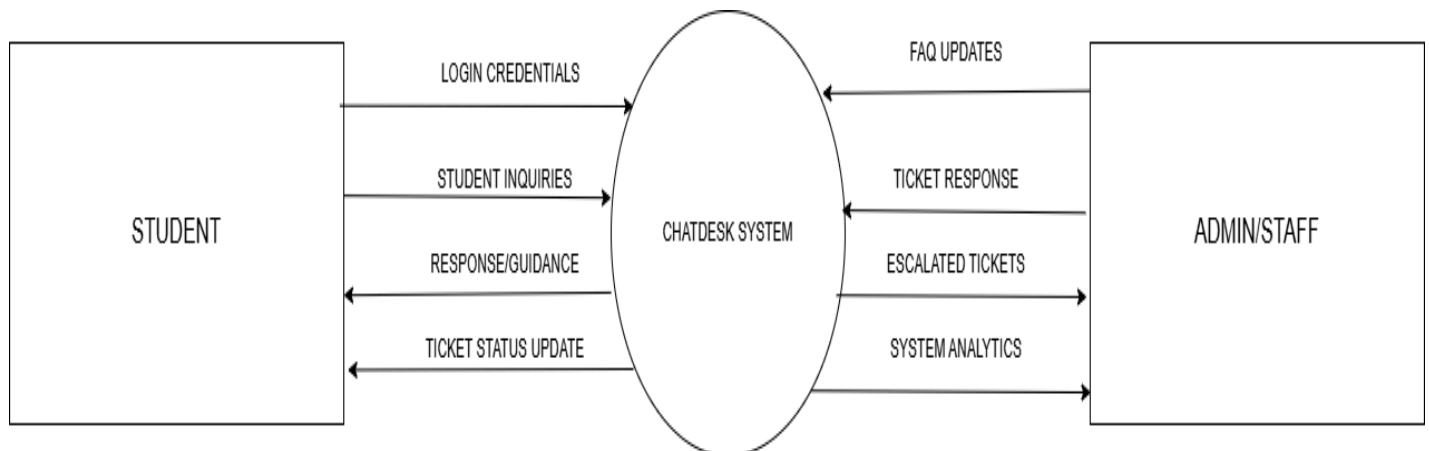


Figure 3. Context Diagram of ChatDesk

3.2.5 Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram illustrates the major internal processes of the ChatDesk system, detailing how information is routed between the primary external entities (Student and Personnel), the core system modules, and the distinct database tables.

The system is comprised of five main processes:

P 1.0 User Authentication: This process handles the secure access for the frontend. The system receives Registration Data or Login Credentials from the STUDENT, checks this data strictly against the unified D1 tbl_user database using User Credential Data, and returns an Authentication Status to grant or deny access.

P 2.0 Inquiry Processing: Acting as the system's AI-powered core, this module receives the text-based Student Inquiry. It processes the natural language to classify the student's intent, sending an FAQ Search Query to D2 tbl_faqs for standard questions. If the AI determines the inquiry is beyond the scope of the knowledge base, it categorizes it as a Complex Inquiry and routes it forward for escalation.

P 3.0 Ticket Escalation: This module catches complex inquiries that the AI cannot automatically resolve. It packages the interaction into a formal support request and stores the Ticket Details securely within the D3 tbl_ticket database for human intervention.

P 4.0 Admin Dashboard: This process serves as the web-based backend interface for university staff. It operates independently from student authentication by sending Admin Credential Data to the D1 tbl_user database. Once logged in, the dashboard acts as a central hub: it retrieves Ticket Records from D3 tbl_ticket for personnel review, pushes FAQ Update Data to D2 tbl_faqs, and compiles system data to store Inquiry Analytics in D4 tbl_inquiry_logs.

P 5.0 Response Generation: This final module handles the system's output back to the student. It formulates the final reply by either retrieving Relevant FAQ Information directly from D2 tbl_faqs (for automated AI replies) or by accepting Staff Resolution Text inputted by the PERSONNEL (for escalated tickets). The formulated Automated Reply or Escalated Response is then sent back to the student's mobile interface.

Data flows from the student to each process, then moves to the respective databases for storage or retrieval, and finally flows back to the student in the form of automated replies, escalated responses, and authentication results.

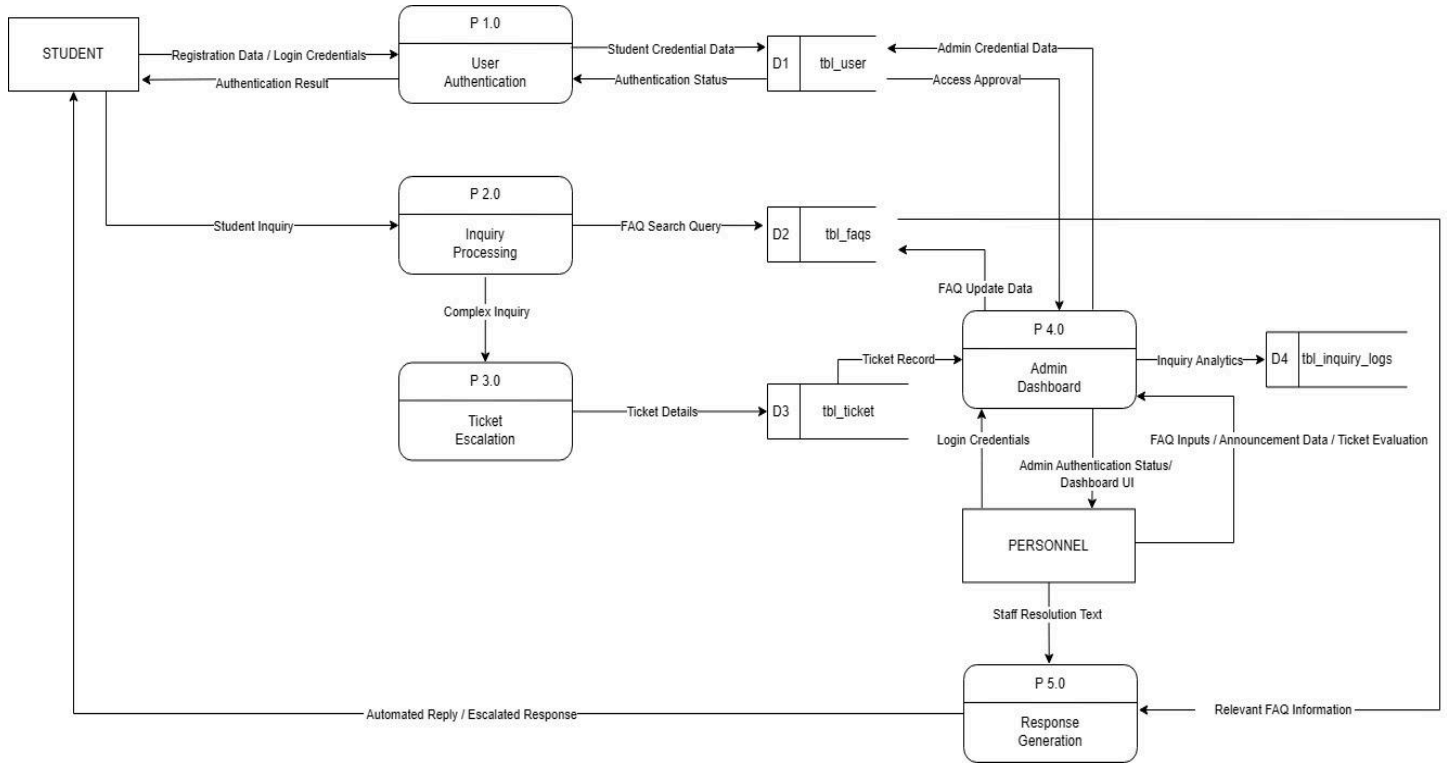


Figure 4. Data Flow Diagram Level 1 of Chatdesk

3.2.6 Level 2 Data Flow Diagram

This Level 2 Data Flow Diagram expands upon the high-level User Authentication process, detailing the specific operational steps required for a student to register an account and securely access the mobile application.

The subsystem is divided into four sub-processes:

P 1.1 User Registration: The STUDENT initiates account creation by submitting their Registration Details. The system requires just a general email for registration, ensuring streamlined onboarding for all users. The process cross-references the unified D1 tbl_user database to check for an Existing User Record to prevent duplicate entries. Upon successful validation, it stores the new User Account Data into the database and issues a Registration Confirmation back to the student.

P 1.2 User Login: For returning users, the student inputs their Login Credentials into the mobile interface. This process securely packages the raw User Credentials and passes them to the verification module.

P 1.3 Credential Verification: This core security module receives the inputted credentials and sends a Credential Validation Request to the D1 tbl_student database. The database returns the corresponding User Account Record. The process then compares the provided credentials against the stored hash to generate a set of Verified Credentials.

P 1.4 Session Access Control: Acting as the final security gate, this process evaluates the Verified Credentials to determine if access should be granted. It generates the final Authentication Result (such as an approved session token or an error prompt for incorrect passwords) and transmits this directly back to the STUDENT, officially granting or denying entry to the application's main features.

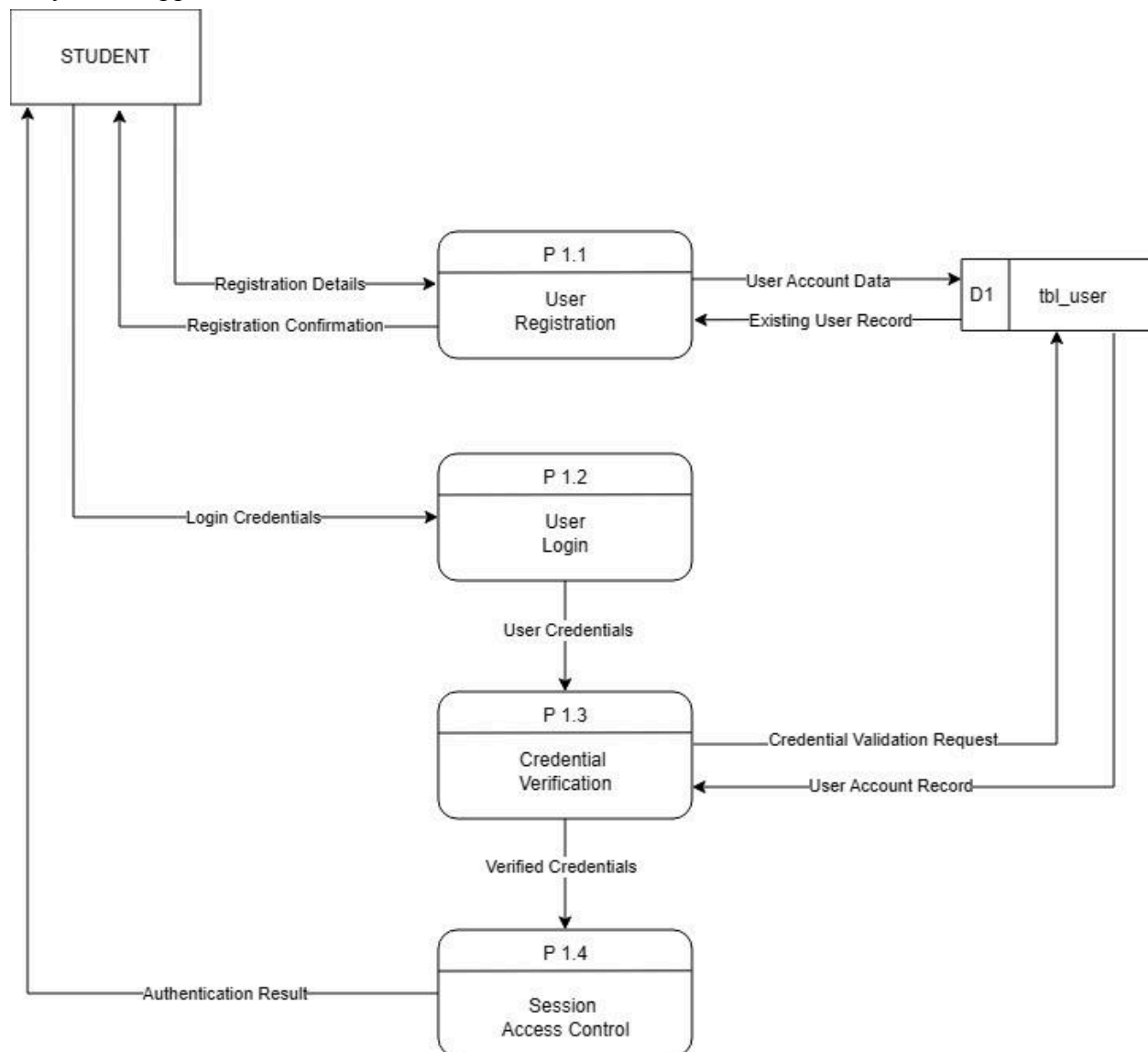


Figure 5. Data Flow Diagram Level 2 of ChatDesk User Authentication

This Level 2 Data Flow Diagram details the AI-powered core of the ChatDesk system, illustrating exactly how incoming messages are analyzed, classified, and routed without human intervention. The subsystem is divided into five sequential sub-processes:

P 2.1 Receive Student Inquiry: The module captures the initial Student Inquiry submitted through the mobile frontend, stripping any unnecessary formatting to extract the Raw Inquiry Text for analysis.

P 2.2 NLP / Intent Classification: Acting as the intelligent engine of the system, this process transmits the raw text to the integrated OpenAI API. The Large Language Model (LLM) processes the natural language—including Taglish phrasing—to accurately determine the user's context and output a Classified Inquiry.

P 2.3 FAQ Matching: Using the AI's classification, this process translates the intent into an explicit FAQ Search Request, which is executed against the D2 tbl_faqs database to search for the university's pre-approved answers.

P 2.4 FAQ Match Validation: The system retrieves the Relevant FAQ Entry from the database and validates the accuracy of the result. It determines whether the system found a highly confident match (Validated FAQ Response) or if the database lacks the appropriate information (Unresolved Inquiry).

P 2.5 Resolution Decision: This final routing module evaluates the validation status. If the query is flagged as an Unresolved Inquiry, the system categorizes it as a Complex Inquiry and forwards the data out to Process 3.0 (Ticket Escalation) for human review. If the query is successfully validated, the system relies on the database to forward the relevant FAQ data to Process 4.0 (Response Generation) to formulate the automated reply.

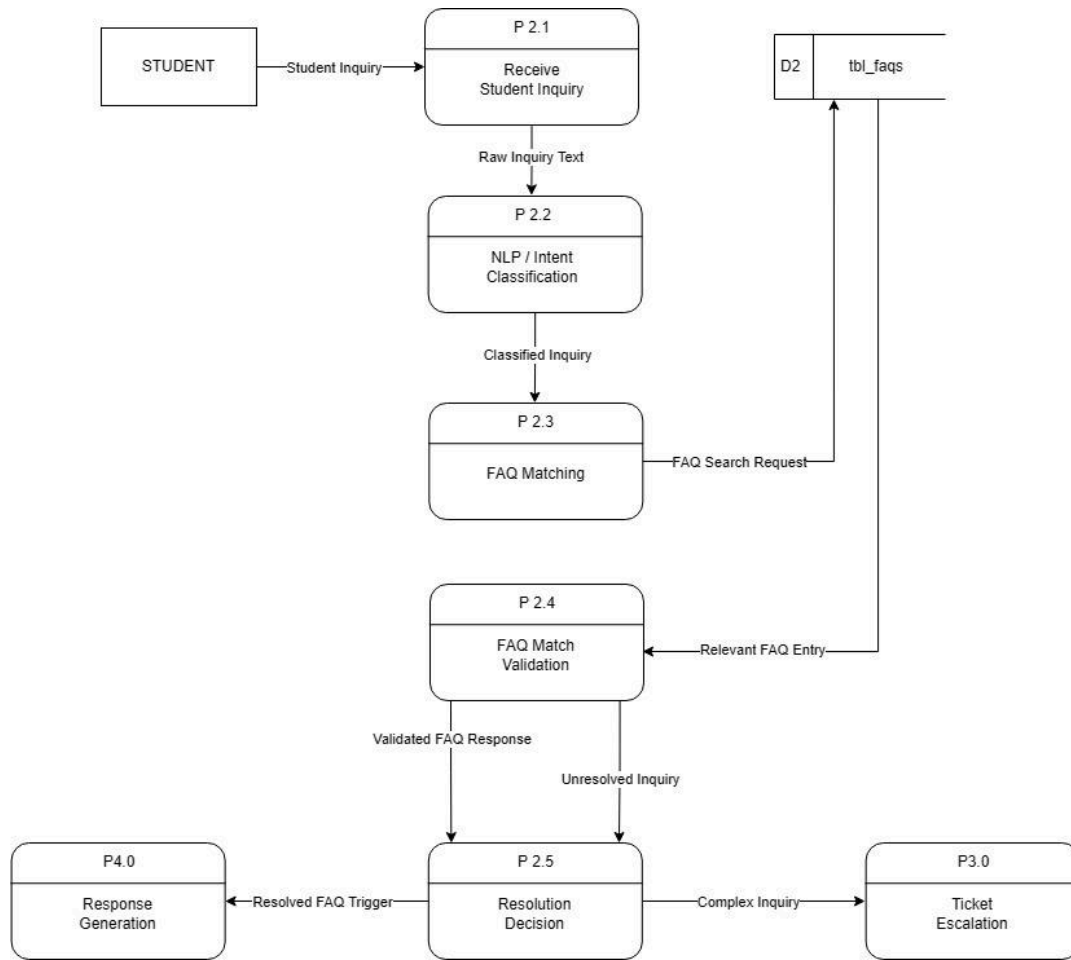


Figure 6. Data Flow Diagram Level 2 of ChatDesk Inquiry Processing

This Level 2 Data Flow Diagram illustrates the automated escalation protocol of the ChatDesk system. This module is triggered exclusively when the AI processing module determines it cannot confidently answer a user's question, ensuring no student concern is left unresolved.

The subsystem consists of three core sub-processes:

P 3.1 Detects Complex Inquiry: This process acts as the intake handler for the escalation module. It receives the routed Complex Inquiry directly from Process 2.0 (Inquiry Processing) after the AI fails to find a valid FAQ match, categorizing the text as an Unresolved Inquiry.

P 3.2 Generate Ticket: The system takes the unresolved inquiry data and packages it into a standardized support format. It compiles the user's intent, the raw message, session data, and the specific log_id directly from the chat history to generate comprehensive Ticket Information.

P 3.3 Store Ticket Information: This final automated step secures the escalated issue. The process executes a database write command, safely depositing the formatted Ticket Details into the D3

tbl_ticket database. Once stored here, the ticket is officially queued and awaits manual retrieval by university staff via the Admin Dashboard.

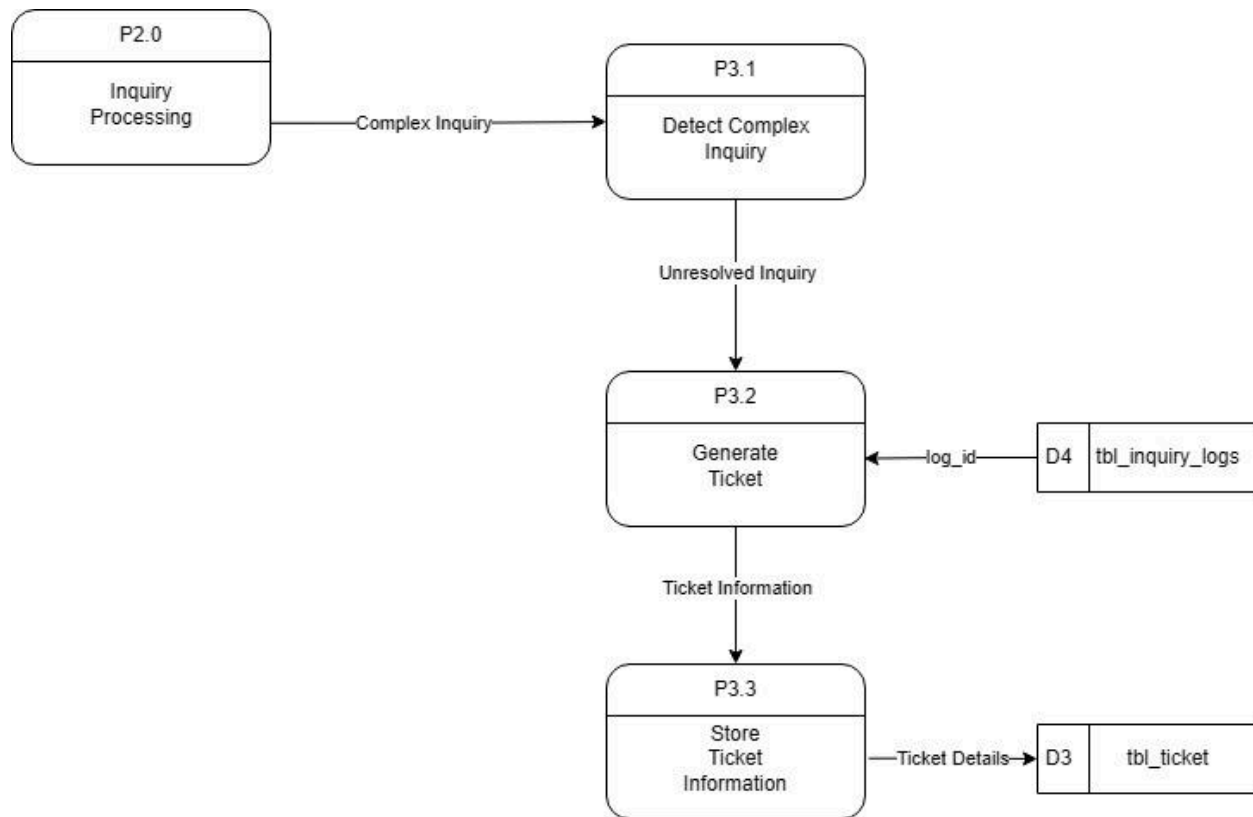


Figure 7. Data Flow Diagram Level 2 of ChatDesk Ticket Escalation

This Level 2 Data Flow Diagram illustrates the backend web interface utilized by the Office of Student Affairs and Services (OSA) personnel. It outlines the administrative tools necessary to maintain the AI chatbot and handle escalated student concerns.

The subsystem is categorized into five administrative processes:

P 5.1 Authenticate Administrator: Ensures secure access to the dashboard. The system receives Login Credentials from the PERSONNEL and strictly validates them against the unified D1 tbl_user database, verifying the specific role attribute before returning a Login Success confirmation upon approval.

P 5.2 Manage Knowledge Base: Allows authorized staff to maintain the AI's accuracy. Personnel input FAQ Updates, which the process directly writes as Updated FAQ Entries into the D2 tbl_faqs database, immediately expanding the chatbot's knowledge.

P 5.3 Manage Announcements: Provides a mechanism for staff to push timely university updates. The system captures Announcement Content and routes it to the central database to be prioritized during student inquiries.

P 5.4 Handle Escalated Tickets: Acts as the manual resolution hub. The process fetches Ticket Records from D3 tbl_ticket, displays them to the personnel for review, and captures the staff's Escalated Ticket Action to send Ticket Updates back to the database.

P 5.5 Analyze Inquiry Trends: Serves as the system's reporting module. Personnel can submit an Analytics Request, prompting the process to retrieve Inquiry Records from D4 tbl_inquiry_logs. The data is then processed and displayed as comprehensive Inquiry Analytics, allowing the university to track peak inquiry times and common student concerns.

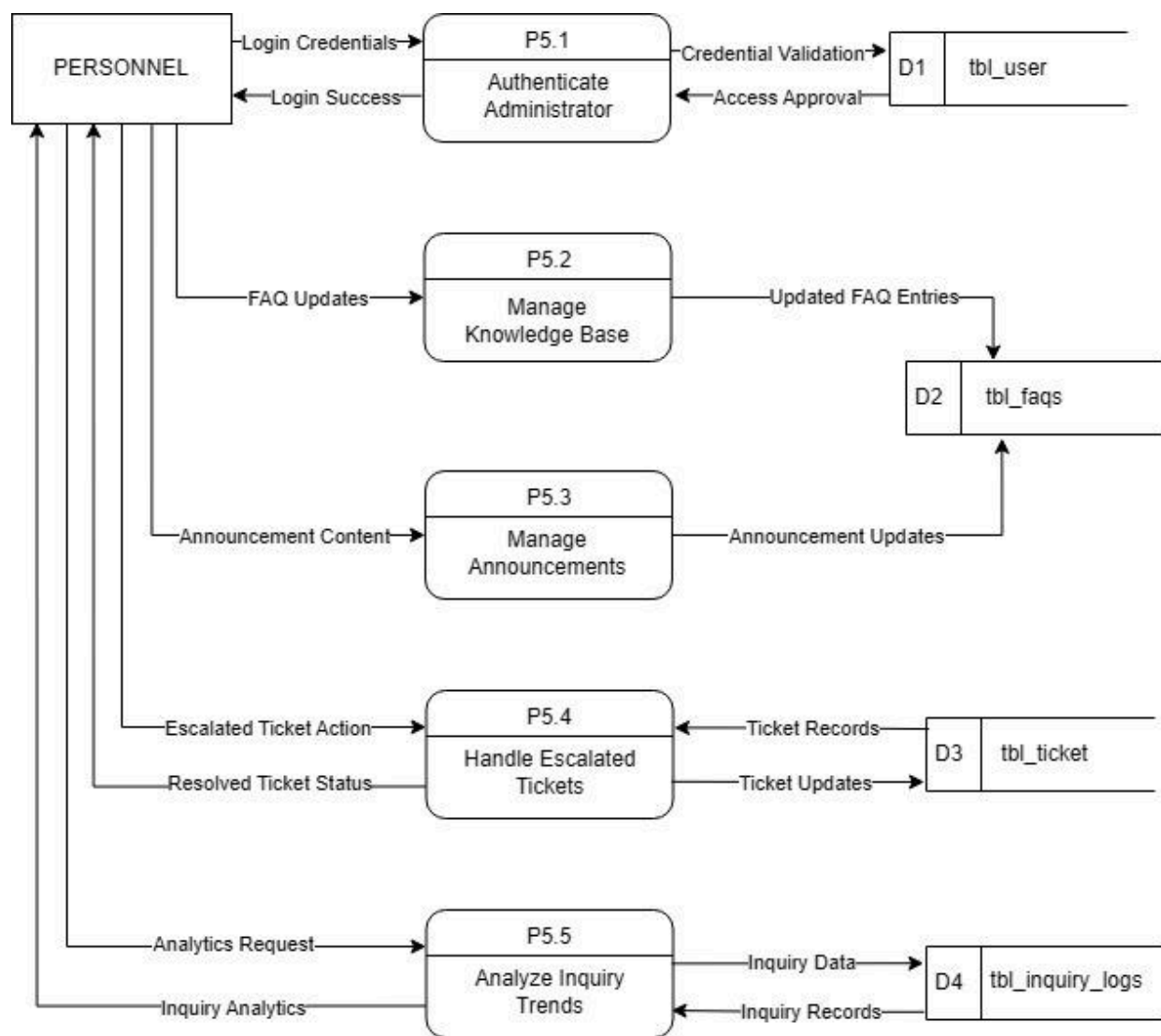


Figure 8. Data Flow Diagram Level 2 of ChatDesk Admin Dashboard

This Level 2 Data Flow Diagram outlines the final communication phase of the ChatDesk system, responsible for formulating and transmitting the final answers back to the mobile interface. To handle both automated and human-assisted resolutions, the subsystem is divided into three distinct sub-processes:

P 4.1 Format Automated Reply: This process handles the AI-driven communication path. Triggered by a successful match in Process 2.0, it retrieves the Relevant FAQ Information directly from the D2 tbl_faqs database. It then structures this data into a conversational, user-friendly automated response.

P 4.2 Format Escalated Reply: This process handles the human-driven communication path. It captures the Manual Response / Resolution typed by the university PERSONNEL after they have reviewed a complex ticket in their dashboard, formatting it for mobile delivery.

P 4.3 Transmit Final Response: Serving as the system's outbound gateway, this final process accepts the formatted messages from either the AI or the staff. It standardizes the data packet and transmits the Automated Reply or Escalated Response directly to the STUDENT, officially closing the inquiry loop.

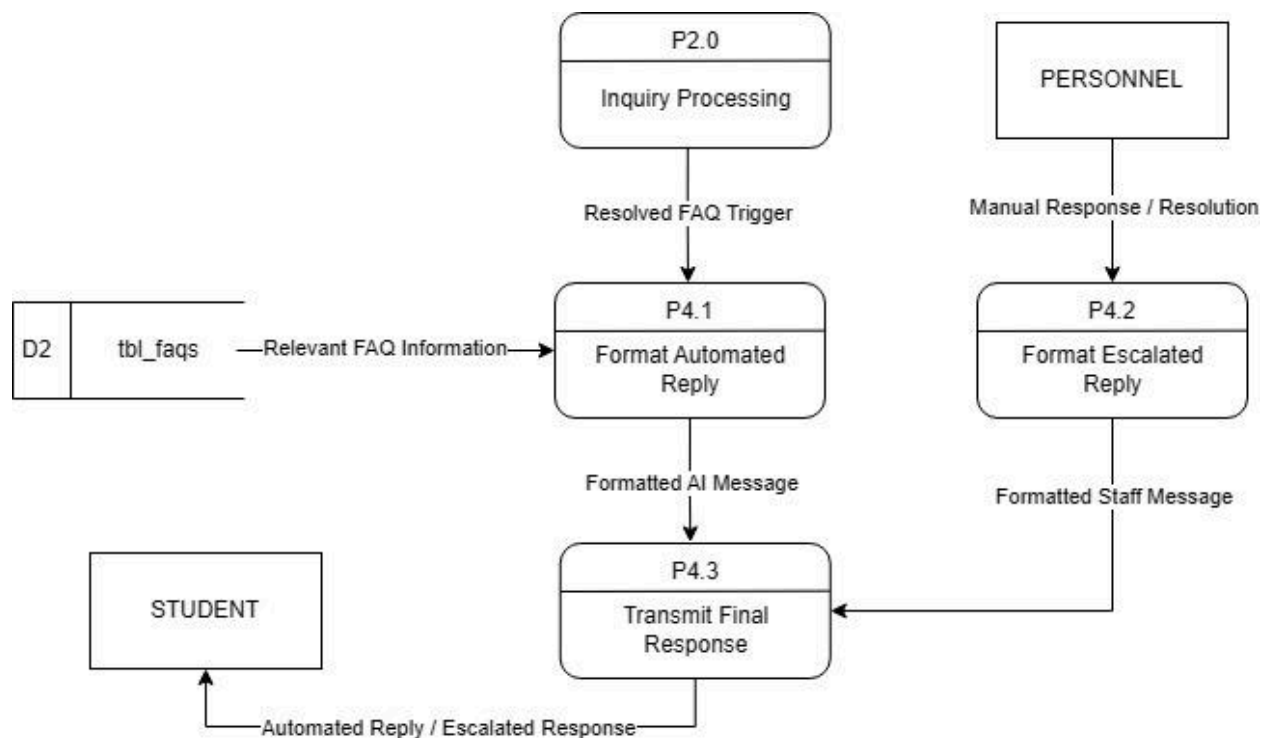


Figure 9. Data Flow Diagram Level 2 of ChatDesk Response Generation

Figure 10 presents the Entity Relationship Diagram of the ChatDesk system. The relational database structure is composed of four core entities: `tbl_user`, `tbl_ticket`, `tbl_inquiry_logs`, and `tbl_faqs`.

The `tbl_user` entity unifies all account information, storing details such as the unique user ID, first and last name, email, password hash, enrolled course, role (to differentiate between students and admins), and account creation timestamps. This entity serves as the central reference for authenticating users from both the mobile application and the administrative web dashboard, and securely linking them to their respective automated chatbot interactions and escalated support tickets.

The `tbl_ticket` entity serves as the core of the hybrid escalation workflow, managing the data for the automatically generated support tickets. It stores the ticket ID, the linked foreign keys for the student (`user_id`), the assigned admin (`resolved_by`), and the specific chat history (`log_id`), as well as the subject category, the full issue description, the current resolution status, and the exact timestamps for when the ticket was created and resolved.

To monitor and evaluate the performance of the artificial intelligence, the `tbl_inquiry_logs` entity records all automated conversations. It stores the user ID alongside the exact user message, the specific intent detected by the natural language processor, the chatbot's generated response, a boolean indicator of whether the chat was escalated (`is_escalated`), and the interaction timestamp.

Lastly, the `tbl_faqs` entity acts as the system's dynamic knowledge base, which is actively managed by the administrators. It stores the intent keyword, the formal question text, the official answer content, the specific departmental category, and administrative tracking data (`created_at`, `updated_at`, and `updated_by`). Overall, this ERD provides a highly normalized database structure that efficiently supports secure user authentication, AI-driven conversational logging, real-time knowledge base management, and a seamless ticket routing workflow within the ChatDesk platform.

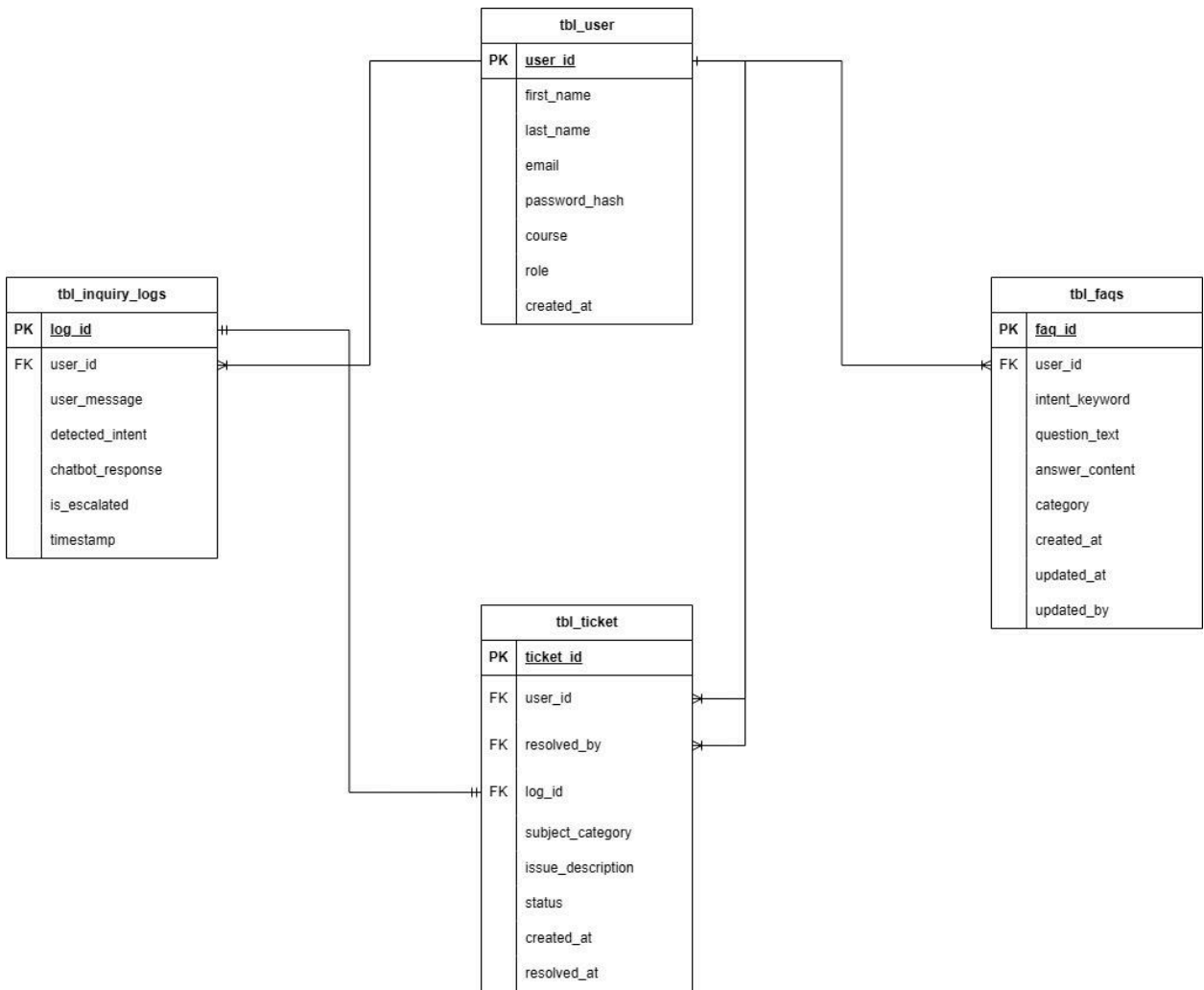


Figure 10. Entity Relationship Diagram of ChatDesk

Hardware Requirement

The hardware requirements for the ChatDesk system are categorized into two phases: the development phase (for the researchers) and the implementation phase (for the end-users and administrators).

Development Phase (Researchers)

To successfully design, code, and test the cross-platform mobile application, web dashboard, and AI integration, the development team requires standard workstation computers with the following minimum specifications:

- **Processor:** Minimum of Intel Core i3 or AMD Ryzen 3
- **Memory (RAM):** Minimum of 8GB (16GB recommended) to efficiently handle Integrated Development Environments (IDEs), database management tools, and React Native mobile emulators.
- **Storage:** At least 256GB SSD to ensure fast read/write speeds when compiling code and managing Node modules or Python environments.
- **Internet Connection:** A stable broadband connection is strictly required for accessing cloud databases, integrating the OpenAI API, and utilizing version control repositories (GitHub).

Implementation Phase (End-Users)

- **Student Users (Mobile App):** Any standard smartphone (Android or iOS) with a basic internet connection (Wi-Fi or Mobile Data). The hardware footprint is minimal, as the heavy AI processing is handled on the cloud backend rather than the student's physical device.
- **OSA Administrators (Web Dashboard):** Standard university-issued office desktop computers or laptops. No specialized hardware is needed; administrators only require a functioning web browser, a keyboard for typing ticket resolutions, and a stable internet connection.
- **Server Infrastructure:** A cloud-based server environment hosted on DigitalOcean to deploy the backend logic, process Natural Language Processing (NLP) API calls, and maintain the PostgreSQL database 24/7. Utilizing this cloud infrastructure ensures

continuous, high-availability access to the ChatDesk system for the student body without relying on a vulnerable, local physical server on the university campus.

Software Requirements

The ChatDesk system utilizes a modern and scalable software stack designed to support real-time conversational AI, natural language processing for Taglish inquiries, hybrid ticket escalation workflows, and administrative helpdesk management. The system includes both mobile and web functionalities to provide students and university organizers with a unified, highly accessible support platform.

To support the multi-platform architecture of ChatDesk, the following software tools and frameworks will be utilized:

Frontend Technologies

- **Mobile Application Development (React Native):** The mobile application is developed using React Native, a cross-platform framework that enables deployment on both Android and iOS devices through a unified codebase. This approach supports rapid development and provides students with a seamless, native-like interface to chat with the AI, automatically escalate unresolved inquiries into support tickets, and track the status of their university inquiries on the go.
- **Web Application Development (React.js):** The web application is developed using React.js, a modern JavaScript library used to create responsive and dynamic single-page applications. It allows the Office of Student Affairs and Services (OSA) frontline staff to efficiently monitor live chat queues, manage the FAQ knowledge base, and resolve escalated support tickets without experiencing page reloads.
- **Frontend Styling (Tailwind CSS):** For interface consistency, Tailwind CSS (Tailwind Labs, 2024) is utilized to create a highly responsive, clean, and modern design specifically for the web-based administrative dashboard.

Backend and Cloud Infrastructure

The backend of the ChatDesk system is developed using the **Django REST Framework**, which provides secure and scalable API services for handling user authentication, ticket routing, chat history logging, and communication between the mobile app, web dashboard, and AI engine. The specific cloud infrastructure includes:

- **DigitalOcean Deployment:** Utilized to host the Django backend, providing a cloud-accessible production API environment. This ensures the chatbot and ticketing system remain online 24/7, independent of local university servers, utilizing scalable cloud infrastructure (DigitalOcean, 2024).
- **Database Management (PostgreSQL):** Serves as the central cloud database for securely storing student accounts, admin credentials, FAQ entries, chat logs, and automated ticket records. PostgreSQL was specifically selected for its advanced ability to efficiently process and store the complex JSON data generated by AI interactions, as well as its superior handling of simultaneous database requests during peak enrollment periods.

Artificial Intelligence Integration

- **Natural Language Processing (OpenAI API):** The core intelligence of the ChatDesk system is driven by the integration of the OpenAI API. The system strictly avoids traditional lexical search mechanisms, which simply scan for hardcoded keywords and fail when students use typographical errors, slang, or complex sentence structures. Instead, ChatDesk utilizes advanced, transformer-based Large Language Models (LLMs) to perform Semantic Intent Classification. This machine learning architecture allows the chatbot to natively process and understand the underlying meaning of complex, unstructured student inquiries in both English and Tagalog, including conversational Taglish. For instance, if a student asks, "Pano mag apply ng scholarship?" or "Ano ang mga requirements para sa financial aid?", a standard database query would fail to connect these drastically different phrases. However, the OpenAI API analyzes the contextual meaning of both messages and standardizes them into a single, recognized intent (e.g., `scholarship_requirements`). It then dynamically maps this classified intent to the exact, correct university procedure stored securely in the PostgreSQL database. This ensures the

system acts as an intelligent conversational filter rather than a glorified search bar, resolving dynamic human language with high accuracy before human escalation is ever required.

Development Environment and Tools

- **Visual Studio Code:** The primary code editor used for all frontend (React) and backend (Django) development.
- **Expo Go:** Enables live mobile testing and rapid iteration for the React Native application.
- **GitHub:** Utilized for version control and collaborative software development among the researchers, ensuring secure codebase management and tracking (GitHub, Inc., 2024).
- **Thunder Client:** A lightweight API testing extension integrated directly into Visual Studio Code. It is utilized for rapid testing and validating API endpoints to ensure secure, real-time data transfer between the Django backend and the frontend interfaces, maximizing development efficiency without the heavy memory consumption of external desktop applications.

3.3 Pilot Testing (Alpha and Beta Testing)

System Architecture and System Workflow

The ChatDesk system follows a 3-tier system architecture composed of three major components: the Presentation Layer (Frontend), the Application Layer (Backend & Cloud), and the Data Layer (Database).

The student-side mobile application (React Native) serves as the primary frontend interface where students can submit inquiries regarding Student Affairs services such as scholarship applications, counseling schedules, shifting procedures, and document requests. Instead of relying solely on basic keyword matching, the application communicates with the backend Application Layer (Django REST Framework), which integrates transformer-based Natural Language Processing (NLP) via the OpenAI API to analyze inquiries and provide accurate, automated responses in real time.

On the administrative side, OSA personnel access the web-based dashboard (ReactJS) to manage frequently asked questions (FAQs), update announcements, monitor inquiries, and

respond to escalated concerns. The backend server acts as the bridge between the chatbot engine, the dashboard, and the Data Layer (PostgreSQL database), ensuring synchronized communication and real-time data logging across the system. If the chatbot cannot automatically resolve a student inquiry, the system seamlessly escalates the concern into a structured support ticket for manual handling by the OSA staff.

Pilot Testing Procedure

Pilot testing will be conducted to evaluate the functionality, usability, and reliability of the ChatDesk system before full implementation. The testing process aims to determine whether the system effectively handles student inquiries and supports Student Affairs operations.

Testing Methodology

- **Alpha Testing (Internal):** Before releasing the system, our development team will conduct Alpha Testing. We will perform white-box and black-box testing using Thunder Client and Expo Go to ensure the Django backend connects properly to the React Native app, and that the OpenAI API accurately routes Taglish queries without crashing
- **Beta Testing (External):** Once the system is stable, we will conduct Beta Testing. We will deploy the application to a controlled group of selected USTP students and offices under Office of Student Affairs and Services(OSA). We will give them specific tasks, like asking the chatbot about scholarships or escalating a ticket, and ask them to report any bugs or incorrect AI responses.

Intended Users

The intended users of the pilot testing include:

- **Personnel from selected offices under the Office of Student Affairs and Services (OSA):** These staff members will evaluate the web-based administrative dashboard, manage the knowledge base, and handle escalated tickets relevant to their specific department.
- **Selected students from the University of Science and Technology of Southern Philippines (USTP):** These students will test the mobile chatbot application by submitting inquiries specifically related to the services handled by the selected OSA offices.

- **System Administrator:** Responsible for overseeing the overall system performance, server stability, and database management during the test phase..

Features to be Tested During Pilot Testing

The following system features will be evaluated during pilot testing:

- User authentication and secure login access
- Accuracy and responsiveness of AI-powered chatbot replies
- Intent classification and FAQ matching capabilities
- Hybrid ticket escalation mechanism
- FAQ knowledge base and announcement management
- Administrative dashboard data retrieval and functionality
- Overall system usability and real-time synchronization

Transactions to be Tested

The following transactions will be performed during pilot testing:

- Submission of mixed-language (Taglish) student inquiries
- Retrieval of automated chatbot responses
- Viewing of announcements and institutional FAQs
- Automated escalation of unresolved or complex inquiries into support tickets
- Updating FAQs and announcements by administrators
- Monitoring, replying to, and closing inquiry tickets via the dashboard

Duration of Pilot Testing

The pilot testing will be conducted for a period of one to two weeks to allow sufficient time for system evaluation, user interaction, and observation of system performance under actual usage conditions. Feedback gathered during the pilot testing period will be analyzed to identify issues and improve the overall functionality and usability of the ChatDesk system before the final system deployment.

3.4 System Performance Evaluation

3.4.1 System Usability Scale (SUS) Testing

The System Usability Scale (SUS), developed by Brooke (1996), will be employed to evaluate the usability, simplicity, efficiency, and overall user experience of the ChatDesk system. The evaluation will focus on determining whether the system effectively assists students in handling inquiries through the AI-powered chatbot and ticket escalation features. The SUS is a standardized 10-item questionnaire that uses a 5-point Likert scale ranging from strongly disagree to strongly agree and is widely used to measure user satisfaction and perceived usability of software systems.

The primary purpose of the performance evaluation is to determine the functionalities and effectiveness of the ChatDesk system. Specifically, the evaluation aims to assess whether the system can provide accurate responses to student inquiries, efficiently escalate unresolved concerns into support tickets, and deliver a user-friendly mobile experience for both students and administrators.

The evaluation will involve the system's intended end-users, specifically selected USTP students and Office of Student Affairs and Services (OSA) personnel, who will assess its usability and functionality based on their actual experience interacting with the application. Additionally, a select group of IT experts will be involved to evaluate the system's technical performance and architectural reliability.

3.4.2 Evaluation Instrument

The researchers will use the **System Usability Scale (SUS)** as the primary evaluation instrument for the system. The SUS is a reliable and widely accepted tool used to evaluate the usability of software applications and information systems. It measures the users' overall perception of the system in terms of usability, ease of use, complexity, and user satisfaction.

The SUS questionnaire consists of 10 standardized statements answered using a **5-point Likert Scale**:

Scale	Verbal Interpretation
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

No	System Usability Scale Question
1.	I think that I would like to use the ChatDesk system frequently.
2.	I found the ChatDesk system unnecessarily complex.
3.	I thought the ChatDesk system was easy to use.
4.	I think that I would need the support of a technical person to be able to use the ChatDesk system.
5.	I found the various functions in the ChatDesk system were well integrated.
6.	I thought there was too much inconsistency in the ChatDesk system.
7.	I would imagine that most people would learn to use the ChatDesk system very quickly.
8.	I found the ChatDesk system very cumbersome to use.

9.	I felt very confident using the ChatDesk system.
10.	I needed to learn a lot of things before I could get going with the ChatDesk system.

The respondents will evaluate the system by answering statements related to the chatbot's responsiveness, ease of navigation, ticket escalation functionality, accessibility, and overall user satisfaction.

3.4.3 Manner of Conducting the Evaluation

The evaluation will be conducted after the development and testing phases of the ChatDesk system are completed. The researchers will first introduce the system to the respondents and provide instructions regarding its features and functions.

The respondents will then be asked to use the mobile inquiry system by performing different tasks such as:

- Sending inquiries through the AI chatbot
- Accessing student service information
- Submitting concerns through the ticket escalation feature
- Viewing chatbot responses and ticket updates

After interacting with the system, the respondents will answer the SUS questionnaire to evaluate their experience using ChatDesk. The accomplished questionnaires will then be collected, tabulated, and analyzed by the researchers.

The evaluation may be conducted through face-to-face testing sessions or online distribution of the questionnaire depending on the availability of the respondents.

3.4.4 Participants / Respondents of the Evaluation

The respondents of the study will strictly consist of the targeted end-users and technical evaluators, categorized as follows:

- **Students of USTP:** Will primarily evaluate the mobile application, focusing on the chatbot's usability, accessibility, and responsiveness in handling inquiries.
- **Office of Student Affairs (OSA) Personnel:** Will assess the web dashboard, focusing on the ease of use regarding ticket escalation, FAQ management, and manual inquiry resolution.
- **IT Experts / System Evaluators:** Will evaluate the technical functionality, interface navigation, and overall system architecture performance.

The selected respondents must have actual experience using the specific modules of the ChatDesk system prior to answering the evaluation questionnaire.

3.4.5 Interpretation of Results Using the SUS Instrument

The responses gathered from the SUS questionnaire will be analyzed using the standard SUS scoring algorithm to determine the overall acceptability and usability of the ChatDesk system. Because of the alternating positive and negative structure of the questions, the score contribution for each item ranges from 0 to 4.

The scores will be calculated using the following rules:

- For odd-numbered items (1, 3, 5, 7, 9), the score contribution is the scale position minus 1 (Score - 1).
- For even-numbered items (2, 4, 6, 8, 10), the score contribution is 5 minus the scale position (5 - Score).
- The sum of the calculated scores is then multiplied by 2.5 to obtain the overall value out of 100.

Formula: Overall SUS Score = (Sum of all item score contributions) * 2.5

The computed SUS scores will be interpreted using the Bangor, Kortum, and Miller (2009) acceptability rating scale:

SUS Score Range	Grade	Adjective Rating	Acceptability
80.3 – 100.0	A	Excellent	Highly Acceptable
68.0 – 80.2	B/C	Good / Okay	Acceptable
51.0 – 67.9	D	Poor	Marginal
0.0 – 50.9	F	Awful	Not Acceptable

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